

Afterlife v1.1 — Graphics, Animation & Teaching Spec

3 October 2026 · Status: awaiting review · Owner: Ujjwal (director) · Builder: Claude · Builds on: 2026-10-03-afterlife-design.md (v1, live at <https://afterlife-one.vercel.app>)

1. Purpose and success criteria

v1 plays well but looks static: a flat dark background, flat plants that “pop” in, and growth animations cut short by hover redraws. It also teaches only through a one-line hint. v1.1 makes the diorama feel **alive** and makes the game **self-explanatory** for a first-time player. The rules, levels, audio and save format do not change.

v1.1 is done when all of these are true:

1. Plants sway gently and grow up from their base. Animations always finish, and hovering never interrupts them.
2. Every player action has clear visual feedback: seed pop, scrap drop with a growth ripple, harvest flight, invalid shake.
3. The scene has atmosphere: a sky gradient that warms with progress, an earth island with a shadow, and drifting particles.
4. Winning has a celebration: a colour wash, fireflies and petals, then the existing camera turn and panel.
5. A first-time player is guided through Bus Stop step by step, and a How to Play screen is reachable from the title and in-game.
6. It is smooth on a mid-range phone, still works with Reduce motion, and the initial download stays under 3 MB compressed.
7. All existing tests still pass, and the new logic has unit tests.

2. Decisions made

Topic	Decision
Rendering approach	Persistent per-tile display objects updated by diff (not a full redraw), plus cached plant textures drawn once per plant, stage and bloom
Upgrades in scope	Living plants, satisfying feedback, atmosphere, celebration and polish (all four)
Teaching	A guided first level on Bus Stop plus a How to Play screen (title button and in-game ?)

Topic	Decision
Unchanged	Engine rules, the 5 levels, audio, save key and format (one new field), and the Kenney sprites

3. Rendering foundation

Layers, back to front: 1. Sky (camera background gradient) 2. Island (the earth slab under the board, with its shadow) 3. Ground tiles 4. Preview (growth-ring outline, placement fill, glow) 5. World: objects and plants sorted by isometric depth 6. Particles 7. Labels (status icons)

Per-tile view. Each grid tile owns a persistent view holding its ground, object and plant display objects. On every controller change the scene computes a **diff** between the previous and new state: - ground tint changed (progress) - object added or removed - plant added, removed, changed stage, or bloom changed - rotation changed, which rebuilds all positions

Only the changed tiles are updated. **The preview layer is redrawn alone on hover**, so it never touches the world layer.

Plant textures. A plant cell is drawn once into a texture keyed by type, stage, bloom, objectHeight and a variant seed, then shown as an image anchored at its base (origin 0.5, 1), and reused. Plant drawings come from an upgraded plantShapes.

Pure logic, unit-tested: - diffTiles(prev, next) → TileChange[] - plantTextureKey(cell, objectHeight) - the animation timing helpers - the tutorial step machine

Phaser-specific code (layers, tweens, particles) is checked in the browser.

4. Visual upgrades

4.1 Living plants

Plant	New look
Moss	Layered clumps: a darker base ellipse, a mid-tone body and a highlight speck
Vine	Curling stems; leaves with a centre vein; on objects, strands wrap up the object
Flower	Leaves with a shine; a bud; on bloom, 6 petals open outward over 0.6 s
Bamboo	Jointed stems with node bands and leaf tufts at the top

- **Grow:** a new stage rises from its base, scaleY 0.2 → 1 over 500 ms, ease Back.out.
- **Spread:** the sprout unfurls, scale 0 → 1 over 450 ms, starting 150 ms after the parent's grow.

- **Sway:** each plant gets a rotation of $\pm(1.5^\circ$ moss, 3° vine, 4° flower, 5° bamboo), a period of 2.4–3.6 s, and a phase seeded from its position. It is driven by one update loop, not one tween per plant.

4.2 Feedback

Action	Animation
Place seed	Mound pops (scale 0.6 $\rightarrow$ 1, 250 ms) plus 4–6 soil specks
Place scrap	Drops from 40 px above with a bounce (350 ms), a dust puff, then a ripple ring expands over exactly its growth tiles (400 ms). The rules apply instantly as before; only each plant's grow <i>animation</i> is delayed until the ripple reaches its tile
Harvest	6 petals burst out, then fly to the seed tray button (600 ms); the tray count bumps
Invalid tap	The red tile shakes horizontally (± 4 px, 200 ms)
Meter	Fill animates (400 ms) and glows for 600 ms when it rises by 5% or more

4.3 Atmosphere

- **Sky:** a vertical gradient from #2b3036 (top) to #3d3a33 (bottom) at 0% progress, lerp to #3b4a3e to #7a6a45 (warm golden) at 100%.
- **Island:** an earth slab 22 px deep under the whole board footprint, with soil layers in three tones, and a soft ellipse shadow (alpha 0.35) underneath.
- **Particles:** 12–20 slow drifting motes. At progress below 0.5 they are grey dust; above it, pollen (pale yellow) and floating seeds.

4.4 Celebration and polish

- **Win:** colour wash from the winning move outwards, tile by tile (40 ms per tile step). Fireflies (14, glowing, wandering for 4 s) and drifting petals. Then the existing camera turn and the “Scene restored” panel.
- **Screen transitions:** a 250 ms fade between title, select, settings, credits, How to Play and levels.
- **Tray icons:** 32 px drawn icons (SVG inline in the HUD) for moss, vine, flower, bamboo and each scrap kind, replacing the colour dot.
- **Reduce motion:** sway, particles, drop and bounce, fly-to-tray and the wash are off; state changes appear instantly. Screen fades become instant.

5. Teaching

5.1 Guided first level (Bus Stop)

It shows when Bus Stop starts and `save.tutorialDone` is false. A hint bubble points at its target and advances only when the player does the action:

Step	Hint	Target	Advances when
1	“Tap Moss in your tray.”	Moss tray button	Moss is selected
2	“Tap a soil tile to plant it.”	A pulsing suggested tile	A seed is placed
3	“Plant one more next to it.”	A suggested neighbouring tile	A second seed is placed
4	“Now pick a Tyre .”	The first tyre tray button	Scrap is selected
5	“Drop it beside your seeds — everything inside the ring grows.”	A suggested tile adjacent to both seeds	Scrap is placed
6	“Keep going! Fill the meter to restore the bus stop.”	The meter	Auto-hides after 4 s or on the next move

- **Skip tutorial** is always visible. Finishing step 6 or skipping sets `tutorialDone = true` (saved).
- **Touch**: the first tap on a tile only previews it. Steps 2, 3 and 5 advance when the item is actually placed (the second tap).
- **Undo during the tutorial** does not move a step backwards. The hint simply stays on the current step.
- **Suggested tiles** are computed from the level, not hard-coded: the first open soil tile near the centre, a neighbour of it, and a scrap tile adjacent to both.

5.2 How to Play screen

- Reachable from a **How to Play** button on the title, the **?** button in the in-game tools, and the Settings screen.
- 5 cards, each with a small inline SVG drawing and one or two sentences:
 1. **Plant**: pick a seed in the tray, then tap a tile.
 2. **Feed**: scrap makes every plant inside its ring grow one step. Small scrap reaches 1 tile, medium 2, large 3.
 3. **Grow**: grown moss and vines spread to new tiles. Flowers bloom, so tap a bloom for a free seed. Bamboo grows tall.
 4. **Restore**: cover the scene, the scrap included, to fill the meter.

5. **Relax:** no timer and no losing. Undo any time; rotate (◀ ▶ or Q/E) and zoom to look around.

- Buttons: **Replay tutorial** (starts Bus Stop with the tutorial) and **Back**. Opened in-game, Back returns to the level and the level state is kept.

5.3 Save format

The existing `afterlife.save.v1` gains `tutorialDone: boolean`, default `false`. Old saves load with `false`.

6. Error handling and performance

- Particles and textures failing** (e.g. a WebGL context loss) must never block play. The world layer still renders, and errors go through the existing error panel only if play is affected.
- Cache size:** the texture cache holds at most about 200 plant textures. Least-recently-used ones are evicted beyond that.
- Frame budget:** 60 fps on desktop; at least 45 fps on a mid-range phone viewport with the Playground fully grown. This is checked in headless Chrome with CPU throttling ×4.
- Download size:** no new runtime dependencies. Icons are inline SVG, particles use Phaser's built-in emitter. The initial download stays under 3 MB compressed.

7. Testing

What	How
<code>diffTiles</code>	Unit tests for each change kind, plus rotation
Plant texture keys and the LRU cache	Unit tests
Animation timing helpers (ripple delay per tile, wash order)	Unit tests
Tutorial step machine, including skip, undo and the touch preview	Unit tests
Suggested tutorial tiles for Bus Stop	Unit test: valid, adjacent, and placeable per the rules
Save <code>tutorialDone</code>	Unit tests: default, round-trip, old save
How to Play and HUD ? / icons	DOM tests (happy-dom)
Visuals, motion, fps, reduce motion, phone	Headless Chrome checks and screenshots of every level and every tutorial step

8. Out of scope

New levels, new plants or mechanics, real-time lighting or shaders beyond Phaser's built-ins, sound changes, and a sandbox or photo mode.